University Saleh Boubnider Constantine 3. Faculty of Medicine. Department of Medicine.
Laboratory of General Anatomy. Academic year 2025-2026.
Course intended for first-year medical students.

GENERAL ANGIOLOGY

PLAN

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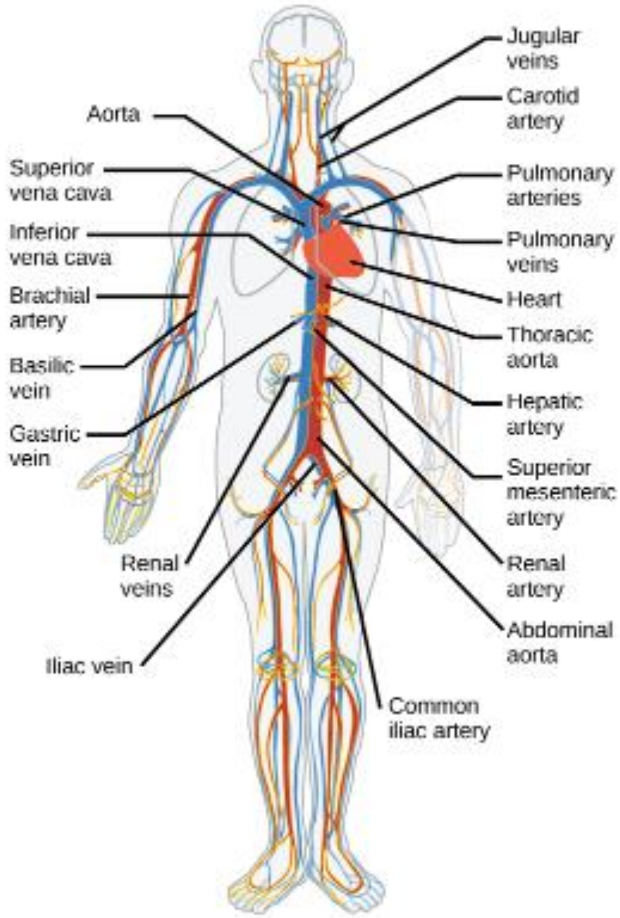
I. INTRODUCTION:

Angiology is the study of the vessels through which a vital liquid circulates throughout the body: blood. It provides tissues with nutrients and oxygen necessary for life and carries back waste products and carbon dioxide. The heart acts as the pump that propels this blood.

• Vessels carrying carbon dioxide and waste products are called veins.
• Vessels carrying oxygen and nutrients are called arteries.
• Vessels carrying lymph are called lymphatic vessels.

In adults, there are about 6 liters of blood in the body. It consists of plasma (around 55%) and cells (around 45%). Plasma is composed of about 90% water and 10% dissolved substances essential for nutrition and tissue function. The cells include red blood cells (erythrocytes) for oxygen and carbon dioxide transport, white blood cells (leukocytes) for defense and protection, and platelets that play a role in blood clotting.

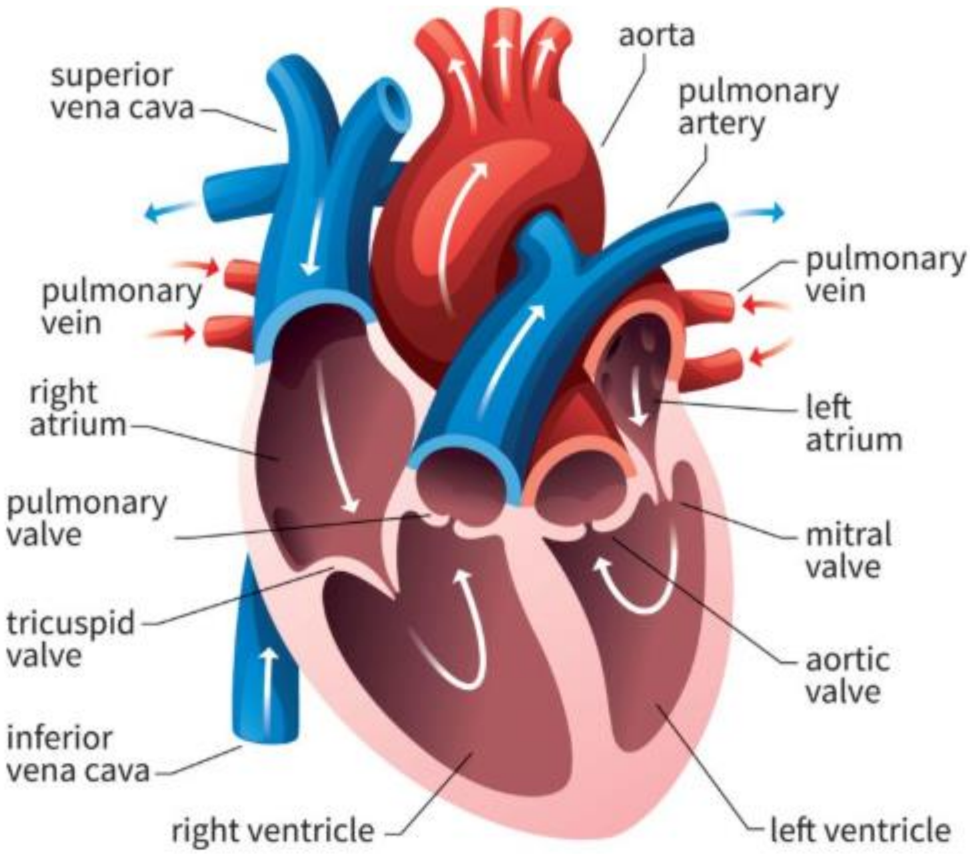


Circulatory system

II. <u>DESCRIPTIVE ANATOMY:</u>

A. <u>THE HEART:</u>

The heart is a hollow fibromuscular organ located in the thoracic cavity between the lungs, resting on the diaphragm. It is the center of the circulatory system, drawing venous blood into its chambers and distributing it to the arteries. The heart is divided into two parts by the interatrial and interventricular septa: the right heart, containing oxygen-poor blood, and the left heart, containing oxygen-rich blood. Each side is subdivided into an atrium and a ventricle, communicating through an atrioventricular opening.



THE HEART

B. <u>THE VESSELS:</u>

Vessels are channels branching throughout the body. They are divided into blood vessels and lymphatic vessels, depending on whether they carry blood or lymph. Blood vessels are classified as arteries or veins.

1. ARTERIES:

Arteries transport blood from the heart to the organs and tissues. They pulsate in sync with heartbeats. The origin of all arteries is the aorta, which arises from the left ventricle and distributes blood to the body via its branches. The pulmonary artery arises from the right ventricle and leads to the lungs. Branches originating before the termination of an artery are called collateral branches, while those from its distal division are terminal branches.

Arteries are found throughout the body except in hyaline cartilage, the cornea, the lens, the epidermis, and skin appendages. Each artery generally has two accompanying veins, except for large arterial trunks.

2. VEINS:

Veins are conduits that return blood to the heart. They contain valves that ensure the centripetal direction of blood flow. According to their position, veins are classified as:

• Deep veins (under the fascia), such as the femoral or azygos veins.

• Superficial veins (above the fascia), visible under the skin, such as those on the back of the hand or the saphenous vein.

3. CAPILLARIES:

Capillaries are very small vessels (5–30 µm) that connect the ends of arteries to the beginnings of veins. They form rich networks where nutrient and gas exchanges occur between blood and tissues.

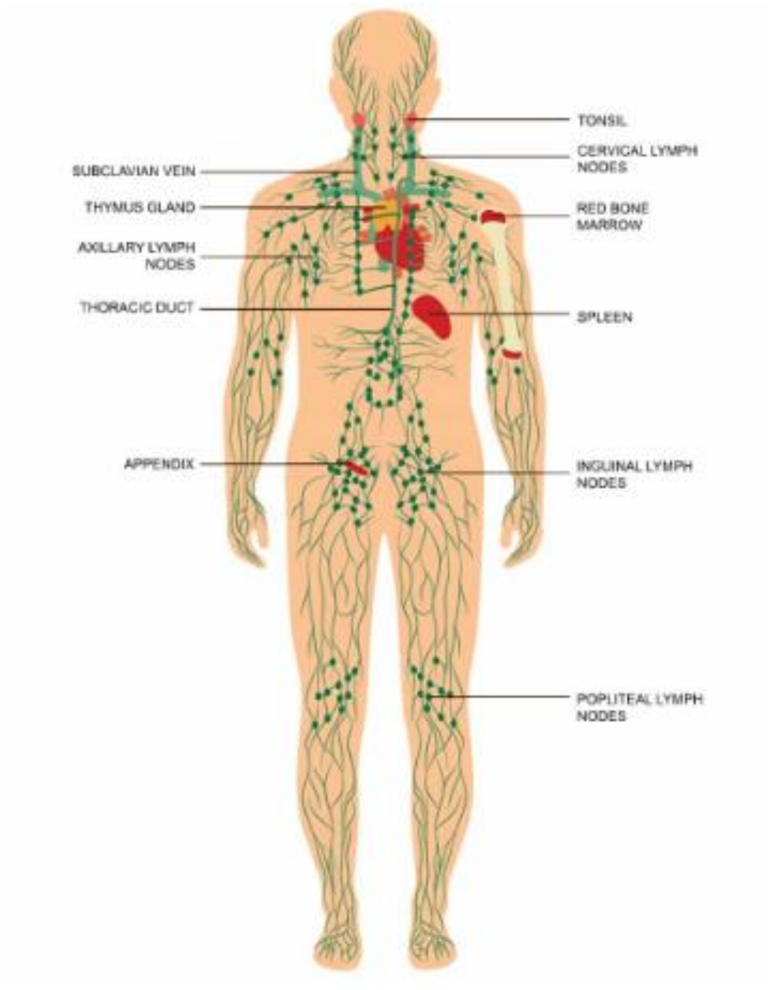
4. THE LYMPHATIC SYSTEM:

The lymphatic system, closely associated with blood vessels, includes lymphatic vessels and nodes, as well as certain organs such as the spleen, thymus (in children), tonsils, and lymphoid follicles found in the respiratory and digestive tracts. Lymphatic vessels carry lymph, a clear yellowish fluid made up mainly of serum and lymphocytes.

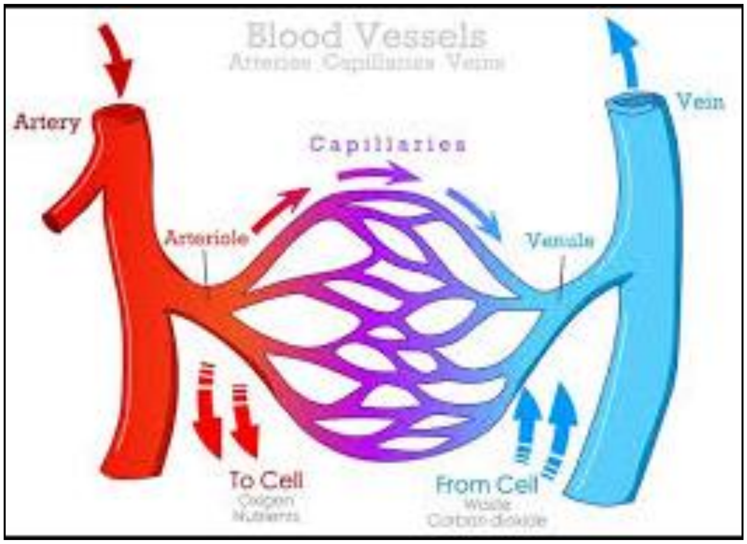
Functions of lymph:

• Nutritional – it transports fats absorbed in the small intestine to the blood.
• Drainage and purification – it carries cellular waste.
• Defense – it transports microbial antigens from infected tissues to lymph nodes.

Lymph nodes are enlargements along lymphatic vessels that filter and purify lymph. About three liters of lymphare drained daily into the venous system.



LYMPHATIC SYSTEM



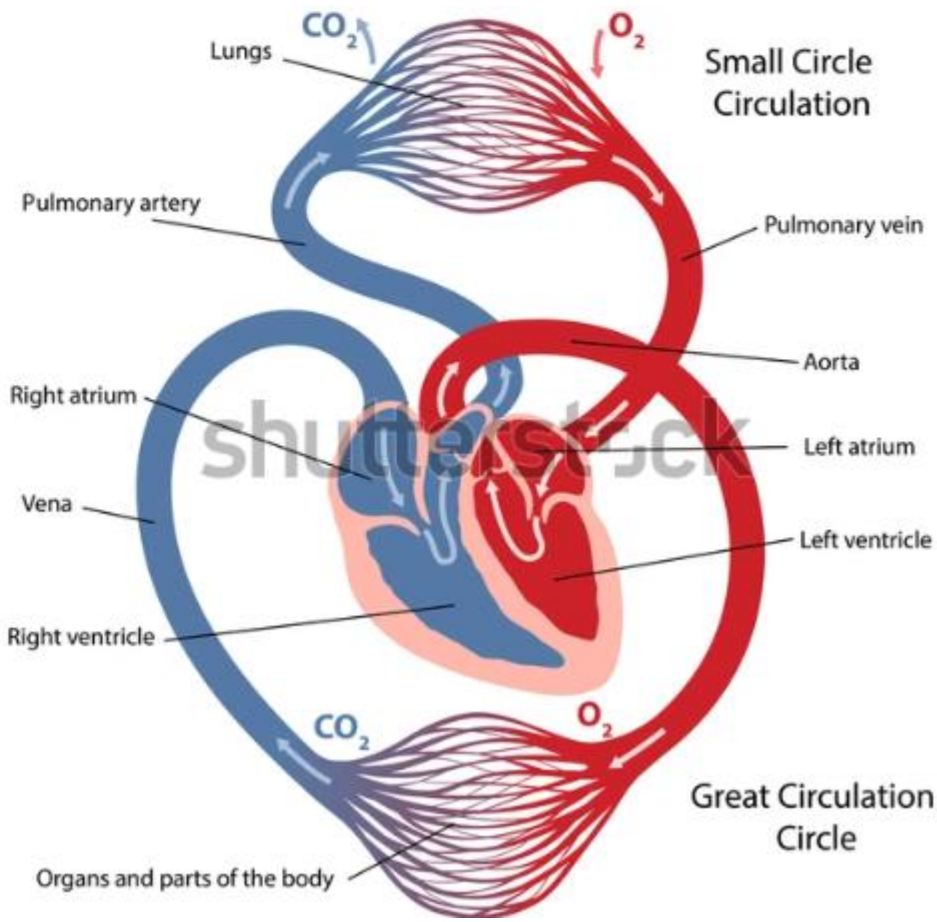
III. <u>THE SMALL (PULMONARY) CIRCULATION:</u>

This circulation ensures hematosis — the transformation of blood rich in carbon dioxide into oxygen-rich blood. It consists of the pulmonary arteries that carry blood to the lungs. The right and left pulmonary arteries arise from the pulmonary trunk, which itself originates from the right ventricle.

Blood from the upper part of the body reaches the heart through the superior vena cava, and that from the lower part through the inferior vena cava. This blood is collected in the right atrium, passes through the tricuspid valve into the right ventricle, and is then ejected into the pulmonary trunk and towards the lungs for oxygenation.

IV<u>. THE LARGE (SYSTEMIC) CIRCULATION:</u>

This circulation, consisting of the aorta, ensures the transport of oxygenated blood to all body tissues. The aorta arises from the base of the left ventricle. Oxygenated blood returns to the heart through the four pulmonary veins, which open into the left atrium. It then passes through the mitral valve into the left ventricle, whose contraction ejects blood into the aorta to distribute it throughout the body.



The small and the large circulation

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